DEPARTMENT OF HEALTH &. HUMAN SERVICES

Public Health Service

Indian Health Service
Phoenix Indian Medical Center
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**Policy Review Statement**
**Reviewer:** Kevin D. Turnage, MSN, RN – Policy and Procedure Officer, Nurse Consultant
**Date:** June 9, 2025
**Policy Title:** Intravenous Electrolytes – Storage, Preparation, and Administration
**PolicyStat ID:** 18096583
**Effective Date:** June 3, 2025
**Next Scheduled Review:** June 2, 2028

## Purpose of Review

This review was conducted to ensure the policy's compliance with federal and accreditation standards, including The Joint Commission (TJC), Centers for Medicare & Medicaid Services (CMS), Indian Health Service Manual (IHM), and FDA regulations. The policy was also assessed for alignment with evidence-based practices, national safety guidelines (ISMP), and the AHA *Get With The Guidelines*® (GWTG) program, as required by PIMC's participation in the Resuscitation Quality Improvement (RQI) initiative.

## Strengths of the Policy

1. **Controlled Access to High-Risk Medications**
   Concentrated electrolytes are restricted to pharmacy storage with limited exceptions, such as magnesium sulfate in emergency carts, consistent with ISMP and TJC safety recommendations for high-alert medications.
2. **Defined Clinical Administration Parameters**
   Clear dosing and infusion parameters are provided for potassium, magnesium, phosphate, and hypertonic sodium, including patient monitoring requirements. These promote standardization and minimize risk of adverse events.
3. **Use of Standardized and Premixed Solutions**
   Promotes pharmacy-prepared or commercial premixed solutions whenever feasible, reducing error risk and aligning with USP <797> for sterile compounding.
4. **Infusion Route and Monitoring Protocols**
   Designates when central venous access is required (e.g., potassium ≥60 mEq/L), with maximum concentration and rate thresholds based on clinical monitoring capacity, enhancing patient safety.
5. **Documentation and Physician Accountability**
   Permits exceptions to standard limits only with physician justification documented in the medical record, reinforcing individualized care with oversight.

## Gaps and Recommendations

1. **Lack of Evidence-Based References**
   The policy lacks citations to authoritative clinical guidance or regulatory frameworks.
   **Recommendation:** Include references to:
   - **ISMP Guidelines** (2022) for high-alert medication safety
   - **FDA labeling and drug safety communications**
   - **AHA ACLS guidelines** for emergency electrolyte use
   - **ACOG or ASHP resources** for electrolyte monitoring and compounding
2. **No Citation of TJC or CMS Standards**
   While "TJC (MM)" is noted in the header, applicable standards are not listed.
   **Recommendation:** Add:
   - **TJC MM.01.01.03** – Safe storage and control of medications
   - **TJC NPSG.03.04.01** – Safe use of high-alert medications
   - **CMS §482.25(b)(1)(iv)** – Safe handling and timely availability of IV drugs
3. **Omission of IHS Manual (IHM) Citations**
   Internal IHS standards governing medication safety are not included.
   **Recommendation:** Cite:
   - **IHM Part 3, Chapter 21** – Pharmacy Services
   - **IHM Part 3, Chapter 1** – Medical Staff responsibilities
4. **No Staff Competency or Training Language**
   The policy does not define staff training or annual validation expectations.
   **Recommendation:** Add:
   - Required annual competency assessment for nurses and pharmacists
   - Initial orientation training on high-risk electrolyte handling and adverse effect recognition
5. **Lacks QA and Reporting Integration**
   The policy does not describe mechanisms for tracking administration incidents or reviewing electrolyte-related events.
   **Recommendation:** Add:
   - Reporting of deviations and errors through **I-STAR**
   - Review of trends by **Pharmacy & Therapeutics** and **Quality Management**

## Integration with AHA Get With The Guidelines® (GWTG) and RQI

As Phoenix Indian Medical Center (PIMC) actively participates in the **Resuscitation Quality Improvement (RQI)** program and supports **AHA's Get With The Guidelines® (GWTG)** initiative, this policy affirms compliance with key clinical performance expectations:

- **GWTG–Heart Failure:** Supports prompt potassium and magnesium repletion to prevent arrhythmias and reduce mortality risk.
- **GWTG–Resuscitation/RQI:** Magnesium sulfate is an ACLS-indicated medication for life-threatening arrhythmias (e.g., torsades de pointes); this policy ensures its safe preparation and availability for code response.
- **GWTG–Stroke:** Encourages appropriate correction of electrolyte abnormalities (e.g., sodium and potassium) which may exacerbate cerebral edema or delay recovery.

This alignment reinforces PIMC's adherence to national cardiovascular and neurologic care benchmarks and supports its RQI accreditation.

## Summary of Recommendations

| Category | Gap | Recommendation |
|---|---|---|
| Evidence-Based Citations | None provided | Add ISMP, FDA, ACLS, AHA references |
| Regulatory Standards | Not listed | Add TJC MM/NPSG and CMS §482.25(b)(1)(iv) |
| IHS Manual Alignment | Omitted | Cite IHM Part 3, Chapters 1 and 21 |
| Training Requirements | Not addressed | Define initial and annual competency |
| Quality Monitoring | No integration | Reference I-STAR and QA committee reviews |

## Conclusion

The *Intravenous Electrolytes – Storage, Preparation, and Administration* policy provides a clinically strong and safety-oriented framework for high-risk IV electrolyte use. To meet full compliance with federal and accreditation standards—and to support PIMC's role in national quality initiatives through GWTG and RQI—targeted revisions are recommended.

**Recommendation:** Maintain the policy as active with revisions to strengthen compliance, evidence-based alignment, and institutional integration.

Respectfully submitted,

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